IEAP RStudio Series 04

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Introduction

This report investigates the influence of random noise on linear regression and explores mathematical models describing heart rate recovery after physical exercise.

The assignment is divided into two main investigations:

1. A numerical experiment examining how different levels of random noise affect linear regression.
2. An analysis of physiological recordings to identify appropriate mathematical models for heart rate recovery.

The analyses combine statistical modelling, graphical representations, and physiological interpretation.

Question 2: Effect of Random Noise on Linear Regression

Question 3: Mathematical Modelling of Heart Rate Recovery

3.1 Scientific Literature and Initial Hypotheses

3.2 Data Exploration and Mathematical Modelling

3.3 Model Comparison and Conclusions

References

The scientific references used throughout this report are listed below.